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Research Paper

The resolution quiver of Nakayama algebras which are minimal Auslander-Gorenstein

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ABSTRACT

Let A be a Nakayama algebra. Using Ringel's resolution quiver, we give a criterion to determine whether A is a minimal Auslander-Gorenstein algebra. The criterion relies strongly on the parity of the selfinjective dimension of A.

1. Introduction

Following Iyama [5], an algebra is said to be higher Auslander provided that its global dimension is finite and bounded by its dominant dimension. Higher Auslander algebras generalize the classical Auslander algebras. Following Iyama and Solberg [6], an algebra is said to be minimal Auslander-Gorenstein provided that its selfinjective dimension is finite and bounded by its dominant dimension. Minimal Auslander-Gorenstein algebras

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are the Gorenstein analogues of higher Auslander algebras. They are a special class of Auslander-Gorenstein algebras.

Nakayama algebras are one of the most fundamental classes of algebras in the representation theory of algebras. Recall that a Nakayama algebra is an algebra for which every indecomposable module is uniserial. Connected Nakayama algebras arise from either linearly oriented quivers or oriented cycles.

The resolution quiver for Nakayama algebras is introduced by Ringel [11]. It is an efficient tool for studying the homological properties of Nakayama algebras. A resolution quiver criterion to detect the finiteness of global dimension and the finiteness of selfinjective dimension for Nakayama algebras is given in [18].

The syzygy filtered algebra for cyclic Nakayama algebras is introduced by Sen [14]. It plays a role in constructing cyclic Nakayama algebras which are higher Auslander [15]. The reversed syzygy filtered algebra for all Nakayama algebras is introduced in [17]. It provides a systematic method to construct cyclic Nakayama algebras which are higher Auslander and minimal Auslander-Gorenstein. A classification of defect one cyclic Nakayama algebras is given in [8], and a classification of concave linear Nakayama algebras which are higher Auslander is given in [13].

Inspired by their work, we study the syzygy filtered algebra for all Nakayama algebras. Then we give a resolution quiver criterion to determine whether a Nakayama algebra is a minimal Auslander-Gorenstein algebra. This criterion depends strongly on the parity of the selfinjective dimension.

Let A be a Nakayama algebra endowed with a cyclic ordering on its vertices. This induces a translation τ on the simple A-modules which agrees with the Auslander-Reiten translation in the cyclic case.

For an A-module M, denote by $\ell(M)$ the composition length, by $P(M)$ the projective cover, and by $I(M)$ the injective envelope of M. As in [11], the resolution quiver $R(A)$ of A is defined as follows. The vertices of $R(A)$ are the isoclasses of all simple A-modules, and there is a unique arrow from S to $\tau \operatorname{soc} P(S)$ for each simple A-module S. A simple A-module S is said to be red if the projective dimension of S is equal to 1, and black otherwise. This assigns a color to each vertex of the resolution quiver.

A simple A-module S is said to be cyclic if S lies on a cycle in $R(A)$, and a leaf if there is no arrow in $R(A)$ ending at S. For any simple A-module S, we denote by $\operatorname{dist}(S)$ the distance between S and the cycles in $R(A)$. This is the minimal integer d such that the d-th successor of S is cyclic. Let $\operatorname{dist}(A)$ be the maximum of $\operatorname{dist}(S)$ over all simple A-modules S.

The weight of the resolution quiver $R(A)$ is the sum of $\ell(P(S))/nc$, where n is the rank of A, c is the number of cycles in $R(A)$, and S runs through all cyclic vertices in $R(A)$.

Resolution quivers serve as an efficient tool for investigating the homological properties of Nakayama algebras. We recall the following result.

Theorem A *([18]). Let A be a Nakayama algebra and $R(A)$ its resolution quiver.*

(1) *A has finite global dimension if and only if $R(A)$ is connected of weight* 1.
(2) *Suppose that A has infinite global dimension. Then A has finite selfinjective dimension if and only if all cyclic vertices in $R(A)$ are black.*

The following is the main result of this paper. It provides a criterion for detecting minimal Auslander-Gorenstein Nakayama algebras purely in terms of combinatorial conditions on the resolution quiver. This criterion relies on the parity of the selfinjective dimension.

Theorem B. *Let A be a Nakayama algebra and $R(A)$ its resolution quiver. Then A is minimal Auslander-Gorenstein of* ***odd*** *selfinjective dimension if and only if*

(1) $\mathrm{dist}(S) = \mathrm{dist}(A) > 0$ *for every leaf S in $R(A)$;*
(2) *every noncyclic vertex has at most* 1 *predecessor in $R(A)$;*
(3) *$R(A)$ is connected of weight* 1*;*
(4) *τ^{-1} of all cyclic vertices in $R(A)$ are black.*

Theorem C. *Let A be a Nakayama algebra and $R(A)$ its resolution quiver. Then A is minimal Auslander-Gorenstein of* ***even*** *selfinjective dimension if and only if*

(1) $\mathrm{dist}(S) = \mathrm{dist}(A) > 0$ *for every leaf S in $R(A)$;*
(2) *every noncyclic vertex has at most* 1 *predecessor in $R(A)$;*
(3) *every cyclic vertex has at most* 2 *predecessors in $R(A)$;*
(4) *all cyclic vertices in $R(A)$ are black.*

We note that condition (3) in Theorem B depends on criterion (1) from Theorem A, while condition (4) in Theorem C generalizes criterion (2) from Theorem A. All remaining conditions are genuinely new.

Throughout this paper, k is a fixed algebraically closed field. All algebras are finite dimensional associative algebras over k. All modules are finite dimensional left modules unless otherwise stated. We refer to [1] for basics on finite dimensional algebras and their representations.

The paper is organized as follows. In Section 2, we recall some basic properties of Nakayama algebras and their resolution quivers. In Section 3, we investigate the syzygy filtered algebra for all Nakayama algebras. Then we prove the main result in Section 4. Some applications and examples are given in Section 5.

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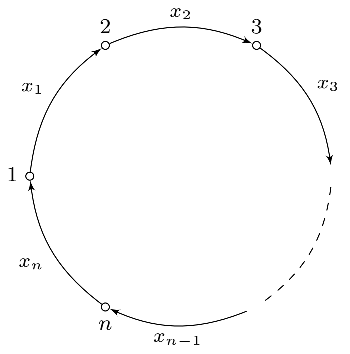


Fig. 1. The quiver C_n.

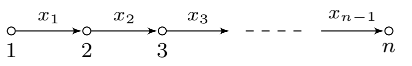


Fig. 2. The quiver L_n.

2. Preliminary

Given an integer n, we denote by C_n the quiver on n vertices containing a single directed cycle through all vertices; see Fig. 1. Denote by kC_n the path algebra of C_n.

Following [4], an algebra A is said to be a Nakayama algebra of rank n if it is isomorphic to the quotient algebra of kC_n via the ideal generated by a finite set I of relations given by nontrivial paths. If I consists of paths of length at least 2, then I is said to be admissible and A is called a cyclic Nakayama algebra.

We allow paths of length one in I. If there is only one of this, say x_n, then A is isomorphic to the quotient algebra of kL_n modulo admissible relations and A is called a linear Nakayama algebra; see Fig. 2. Both cyclic and linear Nakayama algebras are connected. We also allow more than one path of length one in I.

Let A be a Nakayama algebra of rank n. For each vertex a in C_n, we denote by $S(a)$ the simple A-module at a. Set $\tau(a) = b$ if there is an arrow from a to b in C_n. This induces a translation τ on the simple A-modules. If A is a cyclic Nakayama algebra, then τ coincides with the Auslander-Reiten translation [1].

For each vertex a, we write $P(a) = P(S(a))$ and denote by c_a the composition length of $P(a)$. The series

$$c(A) = (c_1, c_2, \cdots, c_n)$$

is called a Kupisch series of A. Two Nakayama algebras are isomorphic if and only if their Kupisch series are equal up to cyclic permutation.

Following [11], the resolution quiver $R(A)$ of A is the successor graph of the function

$$\gamma(S) = \tau \operatorname{soc} P(S)$$

on the set of all isoclasses of simple A-modules; see also [3]. A simple A-module S is said to be red if the projective dimension of S is equal to 1, and black otherwise. A projective A-module P is said to be minimal if its top module is a direct sum of black simple A-modules.

The following is well known.

Lemma 2.1. *Let S be a simple A-module. Then*

(1) *$P(S)$ is injective if and only if $\tau^{-1}S$ is black;*
(2) *$\ell(P(S)) = \ell(P(\tau S)) + 1$ if and only if S is red.*

Corollary 2.2. *Let S be a simple A-module. Then S and $\tau^{-1}S$ are black if and only if $\gamma^{-1}(\gamma S) = \{S\}$.*

We need the following; see [18, 1.1] and [12, 3.10].

Lemma 2.3. *Let A be a Nakayama algebra. Then*

(1) *A has finite global dimension if and only if $R(A)$ is connected of weight 1;*
(2) $2\,\mathrm{dist}(A) - 1 \leq \mathrm{gl.dim}(A) \leq 2\,\mathrm{dist}(A)$, *provided that* $\mathrm{gl.dim}(A)$ *is finite.*

An algebra A is said to be Gorenstein if its left selfinjective dimension and right selfinjective dimension are finite. If A is a Gorenstein algebra, then its left and right selfinjective dimensions are equal; see [19]. It is known that a Nakayama algebra is Gorenstein if and only if its left selfinjective dimension is finite.

We need the following; see [18, 1.2] and [14, 4.13].

Lemma 2.4. *Let A be a Nakayama algebra of infinite global dimension. Then*

(1) *A is Gorenstein if and only if all cyclic vertices in $R(A)$ are black;*
(2) $\mathrm{id}_A A = 2\,\mathrm{dist}(A)$, *provided that* $\mathrm{id}_A A$ *is finite.*

Let A be a Nakayama algebra. A simple A-module S is said to be a leaf if there is no arrow in $R(A)$ ending at S. We need the following; see [7, 3.1].

Lemma 2.5. *Let S be a simple A-module. Then*

(1) *$I(S)$ is projective if and only if τS is a non-leaf;*
(2) $\mathrm{id}_A S = 1$ *if and only if S is a leaf.*

3. Syzygy filtered algebra

In this section, we study the syzygy filtered algebra for all Nakayama algebras. This generalizes Sen's construction for cyclic Nakayama algebras.

Let A be a Nakayama algebra and $R(A)$ its resolution quiver. By [12, A.6] the map γ induces a one-to-one correspondence between the set of all black vertices and the set of all non-leaf vertices in $R(A)$. For any non-leaf vertex S in $R(A)$, let T be the unique black vertex such that $\gamma(T) = S$. Consider the exact sequence

$$0 \to \Delta(S) \to P(\tau T) \xrightarrow{\alpha} P(T) \to T \to 0, \tag{3.1}$$

where $\alpha \in A$ is the arrow starting at T. The kernel $\Delta(S)$ is called a base module. It is the quotient module of $P(S)$ of minimal length such that the translation of its socle is a non-leaf vertex.

For any module M, denote by $\Omega(M)$ the syzygy module of M. Then we have

$$\Delta(S) = \begin{cases} P(S), & \text{if } \operatorname{pd}_A T \in \{0, 2\}, \\ \Omega^2(T), & \text{otherwise.} \end{cases} \tag{3.2}$$

Following [14,16,12], the base set

$$\mathcal{B} = \{\Delta(S) \mid S \text{ is a non-leaf vertex in } R(A)\}$$

form a semibrick in the category $\operatorname{mod} A$ of all finite dimensional A-modules. The filtration category $\mathcal{F}$ of $\mathcal{B}$ is an exact Abelian subcategory of $\operatorname{mod} A$. The category $\mathcal{F}$ contains the projective cover of every base module and the second syzygy of every finite dimensional A-module.

Denote by $\mathcal{P}$ the set of projective covers of all base modules. The syzygy filtered algebra $\varepsilon(A)$ of A is the endomorphism algebra of $\mathcal{P}$, that is,

$$\varepsilon(A) = \operatorname{End}_A(\bigoplus\nolimits_{P \in \mathcal{P}} P).$$

It is a Nakayama algebra of rank less than or equal to the rank of A. The equality holds if and only if A is selfinjective. The category $\operatorname{mod} \varepsilon(A)$ of all finite dimensional $\varepsilon(A)$-modules is equivalent to $\mathcal{F}$. The syzygy filtered algebra $\varepsilon(A)$ has finite global dimension if and only if A has finite global dimension.

Given a Nakayama algebra A, denote by $\chi(A)$ the number of isoclasses of simple projective A-modules. Note that $\chi(A)$ equals the number of occurrences of 1 in the Kupisch series of A. For any integer $i \geq 0$, denote by $\varepsilon^i(A)$ the i-th iterated syzygy filtered algebra of A. By (3.2), we have the following.

Lemma 3.1. *Let A be a Nakayama algebra and $i \geq 1$. Then $\chi(\varepsilon^i(A)) - \chi(\varepsilon^{i-1}(A))$ is equal to the number of isoclasses of simple A-modules of projective dimension $2i$.*

For any $i \geq 1$, the rank of $\varepsilon^i(A)$ is no more than the rank of $\varepsilon^{i-1}(A)$. There is an integer m such that $\varepsilon^m(A)$ is selfinjective. Then $\chi(\varepsilon^m(A))$ is equal to the number of isoclasses of simple A-modules of even projective dimension. This gives an alternative proof of Madsen's criterion [7], which states that a Nakayama algebra A has finite global dimension if and only if there is a simple A-module of even projective dimension.

Following [8], the defect $\operatorname{def} A$ of an algebra A is the number of isoclasses of indecomposable projective non-injective A-modules. If A is a Nakayama algebra, then $\operatorname{def} A$ is equal to the number of red vertices in the resolution quiver of A. By (3.2) we have the following.

Lemma 3.2. *Let A be a Nakayama algebra and $i \geq 0$. Then $\operatorname{def} \varepsilon^i(A)$ is equal to the number of isoclasses of simple A-modules of projective dimension $2i+1$.*

The projective objects in $\mathcal{F}$ are precisely the projective A-modules contained in $\mathcal{F}$. The injective objects in $\mathcal{F}$ are more complicated. Let us give a description.

Let S be a non-leaf vertex in $R(A)$. By [12, A.6] there is a unique vertex T such that $\tau^{-1}T$ is black and $\gamma(T) = S$. Consider the exact sequence

$$0 \to J(\tau^{-1}S) \to P(T) \xrightarrow{w} P(I(\tau^{-1}T)) \to I(\tau^{-1}T) \to 0, \tag{3.3}$$

where $w \in A$ is the concatenation of the maximal nonzero path ending at $\tau^{-1}T$ and the arrow starting at $\tau^{-1}T$.

We have the following; compare [14, 3.18].

Lemma 3.3. *The assignment $S \mapsto J(\tau^{-1}S)$ gives a bijection between the isoclasses of simple A-modules of injective dimension not equal to 1 and the isoclasses of indecomposable injective objects in $\mathcal{F}$. It restricts to a bijection between the isoclasses of simple A-modules of injective dimension 3 and the isoclasses of indecomposable injective non-projective objects in $\mathcal{F}$.*

Proof. If T is a non-leaf vertex, then $I(\tau^{-1}T)$ is projective and w is zero. It follows that $J(\tau^{-1}S)$ is an indecomposable injective A-module contained in $\mathcal{F}$.

If T is a leaf, then $I(\tau^{-1}T)$ is not projective and w is nonzero. Since $P(T)$ is an injective A-module, $J(\tau^{-1}S)$ is an indecomposable object in $\mathcal{F}$. Since the image of w is a proper submodule of a base module, we infer that $J(\tau^{-1}S)$ is an injective object in $\mathcal{F}$.

If S and S' are not isomorphic, the socle modules of $J(\tau^{-1}S)$ and $J(\tau^{-1}S')$ are not isomorphic. Then $J(\tau^{-1}S)$ and $J(\tau^{-1}S')$ are not isomorphic. Note that the number of isoclasses of indecomposable injective objects in $\mathcal{F}$ is equal to the number of non-leaf vertices in $R(A)$. Then the assignment is a bijection.

By Lemma 2.5 $J(\tau^{-1}S)$ is a non-projective object in $\mathcal{F}$ if and only if $\Delta(S)$ is a leaf in $R(\varepsilon A)$. This happens if and only if the injective dimension of S is 3. $\square$

The translation τ induces a translation τ_ε on the simple objects in $\mathcal{F}$. For any vertex $\Delta(S)$ in $R(\varepsilon A)$, by [12, B.4] we have

$$\gamma_\varepsilon(\Delta(S)) = \tau_\varepsilon \operatorname{soc}_{\mathcal{F}} P(\Delta(S)) = \Delta(\gamma S).$$

The assignment $\Delta(S) \mapsto S$ yields an injective map of resolution quivers

$$\varphi\colon R(\varepsilon A) \to R(A).$$

We denote by $R(A)'$ the image of φ. Then $R(A)'$ is obtained from $R(A)$ by removing all leaf vertices. Note that φ preserves the weight. However, φ may not preserve the vertex colors.

A simple A-module S is called $*$-black if $\tau^{-1}S$ is black, and $*$-red otherwise. For any resolution quiver R, denote by Leaf-R, BLeaf-R, and $*$BLeaf-R the set of all leaf vertices, black leaf vertices, and $*$-black leaf vertices, respectively.

For each leaf S in $R(A)'$, there is a unique black leaf T_0 in $R(A)$ such that $\gamma(T_0) = S$, and a unique $*$-black leaf T_1 in $R(A)$ such that $\gamma(T_1) = S$. There is an injective map

$$\varphi_0\colon \text{Leaf-}R(\varepsilon A) \to \text{BLeaf-}R(A)$$

sending $\Delta(S)$ to T_0 and an injective map

$$\varphi_1\colon \text{Leaf-}R(\varepsilon A) \to *\text{BLeaf-}R(A)$$

sending $\Delta(S)$ to T_1. Consider the following two conditions.

(B) For each black leaf in $R(A)$, its successor is a leaf in $R(A)'$.
(B*) For each $*$-black leaf in $R(A)$, its successor is a leaf in $R(A)'$.

Then A satisfies (B) if and only if the map φ_0 is bijective, and A satisfies (B*) if and only if the map φ_1 is bijective.

We have the following.

Proposition 3.4. *Let A be a Nakayama algebra. The following are equivalent.*

(1) *A satisfies (B);*
(2) *φ preserves the vertex colors;*
(3) *the number of red vertices in $R(\varepsilon A)$ and $R(A)'$ is equal;*
(4) *any minimal projective object in $\mathcal{F}$ is a minimal projective A-module.*

Proof. (1) $\iff$ (3) Recall that the number of red vertices is equal to the number of leaf vertices in any resolution quiver. We have

$$|\text{Red-}R(\varepsilon A)| = |\text{Leaf-}R(\varepsilon A)| \leq |\text{BLeaf-}R(A)|$$

and

$$\begin{aligned}|\text{BLeaf-}R(A)| &= |\text{Leaf-}R(A)| - |\text{RLeaf-}R(A)\\ =&|\text{Red-}R(A)| - |\text{RLeaf-}R(A)| = |\text{Red-}R(A)'|.\end{aligned}$$

The equality holds if and only if φ_0 is a bijection.

(2) $\iff$ (3) Let P be an indecomposable projective object in $\mathcal{F}$. If P is minimal projective in $\operatorname{mod} A$, then P is minimal projective in $\mathcal{F}$. This shows that φ preserves red vertices. Then φ preserves the vertex colors if and only if the number of red vertices in $R(\varepsilon A)$ and $R(A)'$ is equal.

(2) $\iff$ (4) This is obvious. $\square$

Similarly, we have the following.

Proposition 3.5. *Let A be a Nakayama algebra. The following are equivalent.*

(1) *A satisfies (B^*);*
(2) *φ preserves the vertex $*$-colors;*
(3) *the number of $*$-red vertices in $R(\varepsilon A)$ and $R(A)'$ is equal;*
(4) *any injective projective object in $\mathcal{F}$ is an injective projective A-module.*

We have the following.

Corollary 3.6. *Let A be a Nakayama algebra satisfying (B). Then A is Gorenstein if and only if $\varepsilon(A)$ is Gorenstein. Furthermore, we have*

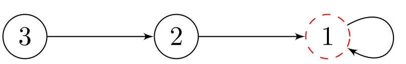
$$\operatorname{id}_A A = \operatorname{id}_{\varepsilon(A)} \varepsilon(A) + 2 \tag{3.4}$$

provided that A is not selfinjective.

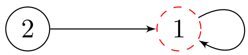
Proof. Since A satisfies (B), then φ preserves the vertex colors. Recall that A has finite global dimension if and only if $\varepsilon(A)$ has finite global dimension. Then by Lemma 2.4 the algebra A is Gorenstein if and only if $\varepsilon(A)$ is Gorenstein.

If $\operatorname{id}_A A = 1$, then A is hereditary by Lemma 2.4. There is a black leaf in $R(A)$. This is impossible. The equation (3.4) follows from [14, 3.23]. $\square$

Example 3.7. Let A be a Nakayama algebra with Kupisch series $(3, 2, 2)$. The resolution quiver $R(A)$ is the following. One checks that A satisfies (B).

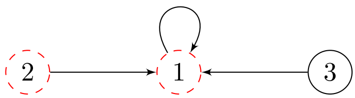


Note that $\Delta(1) = S(1)$ and $\Delta(2) = P(2)$. The Kupisch series of $\varepsilon(A)$ is $(2, 1)$. The resolution quiver of $\varepsilon(A)$ is the following.

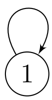


The colors of 1 and 2 are preserved.

Example 3.8. Let A be a Nakayama algebra with Kupisch series $(3, 2, 1)$. The resolution quiver $R(A)$ is the following. One sees that A does not satisfy (B).

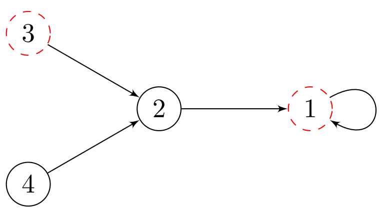


We have $\Delta(1) = P(1)$. The Kupisch series of $\varepsilon(A)$ is (1). The resolution quiver of $\varepsilon(A)$ is the following.



The color of 1 is not preserved.

Example 3.9. Let A be a Nakayama algebra with Kupisch series $(8, 7, 7, 6)$. The resolution quiver of $R(A)$ is the following. It is easy to see that A satisfies (B*).



We have two exact sequences

$$0 \to J(4) \to P(1) \to P(1) \to I(4) \to 0,$$
$$0 \to J(1) \to P(3) \to P(1) \to I(2) \to 0.$$

Note that $J(4)$ is an injective projective A-module, and $J(1)$ is a non-projective A-module.

4. Proof of the main result

In this section, we study the dominant dimension of Nakayama algebras by means of resolution quivers. Then we prove the main result of the present paper.

Let A be an algebra and M be an A-module. Take a minimal injective resolution

$$0 \to M \to I^0 \to I^1 \to \cdots \to I^n \to \cdots$$

of M. The dominant dimension $\operatorname{dom.dim}_A M$ of M is the maximal integer n or ∞ such that I^j is projective for all $j < n$. Dually, take a minimal projective resolution

$$\cdots \to P_n \to \cdots P_1 \to P_0 \to M \to 0$$

of M. The codominant dimension $\operatorname{codom.dim}_A M$ of M is the maximal integer n or ∞ such that P_j is injective for all $j < n$. The left and right dominant dimensions of A are equal; see [9]. It is known that the dominant dimension of any Nakayama algebra is at least 1.

Following [5,6], for an integer $d \geq 1$, an algebra A is called higher Auslander if

$$\operatorname{dom.dim} A = d = \operatorname{gl.dim}(A),$$

and A is called minimal Auslander-Gorenstein if

$$\operatorname{dom.dim} A = d = \operatorname{id}_A A.$$

Higher Auslander algebras are just the minimal Auslander-Gorenstein algebras of finite global dimension.

For algebras of selfinjective dimension 1, we have the following.

Proposition 4.1. *Let A be a Nakayama algebra and $R(A)$ its resolution quiver. Then the selfinjective dimension of A is equal to 1 if and only if*

(1) $\operatorname{dist}(A) = 1$*;*
(2) *$R(A)$ is connected of weight 1;*
(3) *all cyclic vertices in $R(A)$ are $*$-black.*

Proof. “ $\Longrightarrow$ ” Since $\operatorname{id}_A A = 1$, by Lemma 2.4 we have $\operatorname{gl.dim}(A) = 1$. There is a direct sum decomposition of algebras

$$A = A_1 \oplus A_2 \oplus \cdots \oplus A_r,$$

where A_i is a hereditary linear Nakayama algebra for each $i \leq r$. By Lemma 2.3 $\operatorname{dist}(A) = 1$ and $R(A)$ is connected of weight 1. Let S be a cyclic vertex in $R(A)$. Then S is the simple module at the source of A_i and $\tau^{-1}S$ is the simple module at the sink of A_{i-1}. Since $\tau^{-1}S$ is projective, it is black.

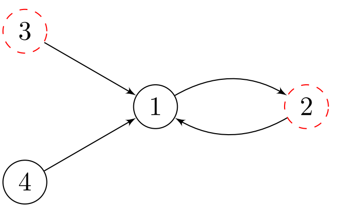
" $\Longleftarrow$ " Let T be a black vertex and S its successor. Since $\operatorname{dist}(A) = 1$, we see that S is cyclic. Then S is $*$-black. By Lemma 2.1 $P(S)$ is injective. By (3.1) there is an exact sequence

$$0 \to \Delta(S) \to P(\tau T) \xrightarrow{\alpha} P(T) \to T \to 0.$$

Since $R(A)$ is connected of weight 1, by Lemma 2.3 the global dimension of A is at most 2. It follows from (3.2) that $\Delta(S) = P(S)$. Then α is zero and T is projective. We conclude that the selfinjective dimension of A is equal to 1. $\square$

Remark 4.2. Let A be a Nakayama algebra of global dimension 1. Since not all connected component of A are simple, there exists at least one black leaf in $R(A)$.

Example 4.3. Let A be a Nakayama algebra with Kupisch series $(1, 3, 2, 1)$. Then A is the direct sum of two hereditary linear Nakayama algebras. The resolution quiver of $R(A)$ is the following.



Note that 4 is a black leaf in the resolution quiver of A.

For algebras of dominant dimension at least 2, we need the following.

Lemma 4.4. *Let A be a Nakayama algebra. The following are equivalent.*

(1) $\operatorname{dom.dim} A \geq 2$*;*
(2) *every leaf in $R(A)$ is $*$-black;*
(3) *every $*$-red vertex in $R(A)$ is the successor of a $*$-black vertex.*

Proof. (1) $\Longrightarrow$ (2) Let S be a $*$-red vertex in $R(A)$. Take a $*$-black vertex T such that $\gamma(T) = \gamma(S)$. Then $P(T)$ is injective. There is an exact sequence

$$0 \to P(S) \to P(T) \to I(\tau^{-1}S).$$

Since $\operatorname{dom.dim} A \geq 2$, we have $I(\tau^{-1}S)$ is projective. Then S is not a leaf.

(2) $\Longrightarrow$ (3) Let S be a $*$-red vertex in $R(A)$. Since S is not a leaf, there is a $*$-black vertex T such that $\gamma(T) = S$. Then S is the successor of a $*$-black vertex.

(3) $\implies$ (1) This follows from [10, 6.2]. Let P be an indecomposable projective non-injective A-module. Then its top module S is $*$-red and there is a $*$-black simple A-module T such that $\gamma(T) = S$. Since T is $*$-black, $P(T)$ and $I(\tau^{-1}S)$ are isomorphic. It follows that $\operatorname{dom.dim} A \geq 2$. $\square$

We have the following.

Proposition 4.5. *Let A be a Nakayama algebra and $R(A)$ its resolution quiver. Then A is minimal Auslander-Gorenstein of selfinjective dimension* 2 *if and only if*

(1) $\operatorname{dist}(A) = 1$*;*
(2) $|\gamma^{-1}(S)| \leq 2$ *for every cyclic vertex S in $R(A)$;*
(3) *all cyclic vertices in $R(A)$ are black.*

Proof. " $\implies$ " (1) If A has finite global dimension, then $\operatorname{dist}(A) = 1$ by Lemma 2.3. If A has infinite global dimension, by Lemma 2.4 we have $\operatorname{dist}(A) = 1$.

(2) Suppose three distinct vertices $T, \tau T, \tau^2 T$ have the same successor in $R(A)$. We may assume that T and τT are red. Since $\operatorname{dom.dim} A \geq 2$, by Lemma 4.4 τT and $\tau^2 T$ are non-leaf vertices. Since $\operatorname{dist} A = 1$, both τT and $\tau^2 T$ are cyclic. This is impossible.

(3) Let S be a red vertex in $R(A)$. By (2) the vertex $\tau^{-1}S$ is black. Then $P(S)$ is injective. Since $\operatorname{id}_A A = 2$, there is an exact sequence

$$0 \to P(\tau S) \to P(S) \to I(S) \to I(\tau^{-1}S) \to 0.$$

By Lemma 2.5 S is a leaf in $R(A)$. In particular, S is not cyclic.

" $\impliedby$ " Since all cyclic vertices in $R(A)$ are black, by Lemma 2.4 the algebra A is Gorenstein. Since $\operatorname{dist}(A) = 1$, the selfinjective dimension of A is at most 2. By Remark 4.2 A is not hereditary. Then the selfinjective dimension of A is 2.

Let S be a leaf vertex in $R(A)$. Since $\operatorname{dist}(A) = 1$ and all cyclic vertices are black, then S is red and $\gamma(S) = \gamma(\tau S)$. Since $\gamma(S)$ has at most 2 distinct predecessors, $\tau^{-1}S$ is black. By Lemma 4.4, the dominant dimension of A is at least 2. Then A is minimal Auslander-Gorenstein of selfinjective dimension 2. $\square$

Example 4.6. Let A be a Nakayama algebra with Kupisch series $(6, 7, 6, 7)$. The resolution quiver $R(A)$ is the following.

$$\begin{tikzcd} 2 \arrow[r] & 1 \arrow[r, bend left] & 3 \arrow[l, bend left] & 4 \arrow[l] \end{tikzcd}$$

It is easy to check that A fulfills the conditions in Proposition 4.5. Then A is a minimal Auslander-Gorenstein algebra of selfinjective dimension 2.

For algebras of dominant dimension at least 3, recall the following key result from [17, 4]. We give a short proof here.

Lemma 4.7. *Let A be a Nakayama algebra. The following are equivalent.*

(1) $\operatorname{dom.dim} A \geq 3$*;*
(2) *every leaf in $R(A)$ is black and $*$-black;*
(3) $\gamma^{-1}(\gamma S) = \{S\}$ *for every leaf S in $R(A)$;*
(4) $\operatorname{def} A = \operatorname{def} \varepsilon(A)$*;*
(5) $\operatorname{dom.dim} A = \operatorname{dom.dim} \varepsilon(A) + 2$*.*

Proof. (1) $\Longrightarrow$ (2) Let S be a leaf in $R(A)$. Since $\operatorname{dom.dim} A \geq 2$, by Lemma 4.4 S is $*$-black and $P(S)$ is injective. If S is red, there is an exact sequence

$$0 \to P(\tau S) \to P(S) \to I(S) \to I(\tau^{-1}S) \to 0.$$

Since $\operatorname{dom.dim} A \geq 3$, then $I(\tau^{-1}S)$ is projective. This is impossible. We conclude that S is black and $*$-black.

(2) $\Longleftrightarrow$ (3) This follows from Corollary 2.2.

(3) $\Longleftrightarrow$ (4) Note that $\operatorname{def} \varepsilon(A)$ is equal to the number of leaf vertices in $R(A)'$. Then $\operatorname{def} A$ and $\operatorname{def} \varepsilon(A)$ are equal if and only if every leaf in $R(A)$ is the unique predecessor of its successor.

(2) $\Longrightarrow$ (5) Let T be a leaf in $R(A)$ and S the successor of T. Since $\tau^{-1}T$ is black, by (3.3) there is an exact sequence

$$0 \to J(\tau^{-1}S) \to P(T) \xrightarrow{w} P(I(\tau^{-1}T)) \to I(\tau^{-1}T) \to 0.$$

Since T is a leaf, we have $w \neq 0$. Note that A satisfies (B*). By Proposition 3.5 any projective injective object in $\mathcal{F}$ is a projective injective A-module. Then

$$\operatorname{codom.dim}_A I(\tau^{-1}T) = \operatorname{codom.dim}_{\mathcal{F}} J(\tau^{-1}S) + 2.$$

By Lemma 3.3 we have $\operatorname{dom.dim} A = \operatorname{dom.dim} \varepsilon(A) + 2$.

(5) $\Longrightarrow$ (1) Since $\operatorname{dom.dim} \varepsilon(A) \geq 1$, we have $\operatorname{dom.dim} A \geq 3$. $\square$

Note that the condition $\operatorname{dom.dim} A \geq 3$ implies both (B) and (B*). We have the following.

Corollary 4.8. *Let A be a Nakayama algebra of dominant dimension at least 3 and $\Delta(S)$ a vertex in $R(\varepsilon A)$. Then*

(1) $\Delta(S)$ *is black if and only if S is black;*
(2) $\Delta(S)$ *is $*$-black if and only if S is $*$-black.*

Combining Lemma 4.4, Lemma 4.7 and Corollary 4.8, we obtain the following.

Corollary 4.9. *Let A be a Nakayama algebra and $m \geq 0$.*

(1) *The following are equivalent.*
  (a) $\operatorname{dom.dim} A \geq 2m+1$;
  (b) $S, \gamma(S), \cdots, \gamma^{m-1}(S)$ *are black and* $*$*-black for each leaf* S;
  (c) $|\gamma^{-1}(\gamma^i S)| = 1$ *for each* $i \leq m$ *and each leaf* S.

(2) *The following are equivalent.*
  (a) $\operatorname{dom.dim} A \geq 2m+2$;
  (b) $S, \gamma(S), \cdots, \gamma^{m-1}(S)$ *are black and* $*$*-black, and* $\gamma^m(S)$ *is* $*$*-black for each leaf* S;
  (c) $|\gamma^{-1}(\gamma^i S)| = 1$ *and* $\gamma^m(S)$ *is* $*$*-black for each* $i \leq m$ *and each leaf* S.

By Lemma 4.7 and Corollary 3.6, we have the following; compare [17, 4.4.2].

Proposition 4.10. *Let A be a Nakayama algebra and $d \geq 3$. Then A is minimal Auslander-Gorenstein of selfinjective dimension d if and only if A has dominant dimension at least 3 and $\varepsilon(A)$ is minimal Auslander-Gorenstein of selfinjective dimension $d-2$.*

Now we restate and prove Theorem B.

Theorem 4.11. *Let A be a Nakayama algebra and $m \geq 1$. Then A is minimal Auslander-Gorenstein of selfinjective dimension $2m-1$ if and only if*

(1) $\operatorname{dist}(S) = \operatorname{dist}(A) = m$ *for every leaf S in $R(A)$;*
(2) $|\gamma^{-1}(S)| \leq 1$ *for every noncyclic vertex S in $R(A)$;*
(3) *$R(A)$ is connected of weight 1;*
(4) *all cyclic vertices in $R(A)$ are $*$-black.*

Proof. “ $\Longrightarrow$ ” We prove by induction on m. If $m=1$, this is done in Proposition 4.1. Suppose $m \geq 2$. By Proposition 4.10 the syzygy filtered algebra $\varepsilon(A)$ is minimal Auslander-Gorenstein of selfinjective dimension $2m-3$. By induction hypothesis, the following statements hold.

(1a) $\operatorname{dist}(\Delta) = \operatorname{dist}\varepsilon(A) = m-1$ for every leaf Δ in $R(\varepsilon A)$;
(2a) $|\gamma_\varepsilon^{-1}(\Delta)| \leq 1$ for every noncyclic vertex Δ in $R(\varepsilon A)$;
(3a) $R(\varepsilon A)$ is connected of weight 1;
(4a) all cyclic vertices in $R(\varepsilon A)$ are $*$-black.

Since there is an injective weight preserving map of resolution quivers from $R(\varepsilon A)$ to $R(A)$, then (1), (2) and (3) hold. By Corollary 4.8 the statement (4) holds.

" $\Longleftarrow$ " We prove by induction on m. If $m = 1$, this is shown in Proposition 4.1. Suppose $m \geq 2$ and let S be a leaf in $R(A)$. Since $\mathrm{dist}(S) \geq 2$, the successor of S is not cyclic. Then S is the unique predecessor of its successor. By Lemma 4.7 the dominant dimension of A is at least 3.

Since there is an injective weight preserving map of resolution quivers from $R(\varepsilon A)$ to $R(A)$, then (1a), (2a) and (3a) hold. By Corollary 4.8 the statement (4a) holds.

By the induction hypothesis, the algebra $\varepsilon(A)$ is minimal Auslander-Gorenstein of selfinjective dimension $2m-3$. Since the dominant dimension of A is at least 3, by Proposition 4.10 the algebra A is minimal Auslander-Gorenstein of selfinjective dimension $2m-1$. $\square$

Let us restate Theorem C as follows. The proof is similar, we omit it here.

Theorem 4.12. *Let A be a Nakayama algebra and $m \geq 1$. Then A is minimal Auslander-Gorenstein of selfinjective dimension $2m$ if and only if*

(1) $\mathrm{dist}(S) = \mathrm{dist}(A) = m$ *for every leaf S in $R(A)$;*
(2) $|\gamma^{-1}(S)| \leq 1$ *for every noncyclic vertex S in $R(A)$;*
(3) $|\gamma^{-1}(S)| \leq 2$ *for every cyclic vertex S in $R(A)$;*
(4) *all cyclic vertices in $R(A)$ are black.*

For a Nakayama algebra A, denote by $\mathrm{cyc}\, A$ the number of cyclic vertices in its resolution quiver.

Corollary 4.13. *Let A be a Nakayama algebra. If A is a minimal Auslander-Gorenstein algebra of even selfinjective dimension, then* $\mathrm{def}\, A \leq \mathrm{cyc}\, A$.

5. Applications and examples

In this section, we present some applications and examples to illustrate the criteria obtained in Theorem B and Theorem C. These examples and applications demonstrate how the resolution quiver can be used to detect minimal Auslander-Gorenstein Nakayama algebras.

Let A be a Nakayama algebra with Kupisch series

$$(c_1, c_2, \cdots, c_n).$$

Denote by $\bar{A}$ the Nakayama algebra with Kupisch series

$$(c_1 + n, c_2 + n, \cdots, c_n + n).$$

It is straightforward to show that there is a color preserving isomorphism between $R(\bar{A})$ and $R(A)$. However, the weight of $R(\bar{A})$ is greater than the weight of $R(A)$.

By Theorem C, we have the following.

Proposition 5.1. *Let A be a Nakayama algebra and d an even number. Then A is minimal Auslander-Gorenstein of selfinjective dimension d if and only if so is $\bar{A}$.*

Let $2 \leq m \leq n$. Denote by $\mathbb{A}_n$ the hereditary linear Nakayama algebra of rank n. Let $N_n(m)$ be the quotient algebra of $\mathbb{A}_n$ via the ideal generated by all paths of length m. The algebra $N_n(m)$ is a linear Nakayama algebra and has finite global dimension. The Kupisch series of $N_n(m)$ is

$$((m)^{n-m+1}, m-1, \cdots, 2, 1).$$

If $m = 2$, then $N_n(m)$ is higher Auslander of global dimension $n-1$. Suppose $m > 2$ and $N_n(m)$ is a higher Auslander algebra. We write

$$n = qm + r$$

such that q, r are integers and $0 \leq r \leq m-1$. Since $S(1)$ is cyclic and $S(1)$ has m predecessors in the resolution quiver, the global dimension of $N_n(m)$ is odd by Theorem C. Since $S(n-r+1)$ is cyclic, $S(n-r)$ is black by Theorem B. It follows that $r = 0$ and m divides n.

If m divides n, it is routine to check that $N_n(m)$ is higher Auslander of global dimension $2n/m - 1$.

Using the resolution quiver criterion, we give an alternative proof of Proposition 6.8 in [2].

Proposition 5.2. *The algebra $N_n(m)$ is a higher Auslander algebra if and only if $m = 2$ or m divides n. In both cases, the global dimension of $N_n(m)$ is $2n/m - 1$.*

Let A be a Nakayama algebra. Following [17], there exists a unique Nakayama algebra $\varepsilon^{-1}(A)$ of dominant dimension at least 3 such that $\varepsilon(\varepsilon^{-1}(A))$ is naturally isomorphic to A. We recall that this algebra $\varepsilon^{-1}(A)$ is defined as follows.

Suppose the Kupisch series of A is

$$c(A) = (c_1, c_2, \cdots, c_n).$$

Denote by d_a the maximum of $c_a - c_{a-1}$ and 0 for each $a \leq n$, and by d the sum of all d_a. The subscript is taken modulo n. Let $\tilde{c}_a$ be a series of length $1 + d_{a+1}$ with

$$(\tilde{c}_a)_\ell = c_a + \sum_{i=1}^{c_a+\ell-1} d_{a+i}, \quad \text{for each } \ell \leq 1 + d_{a+1}.$$

Then $\varepsilon^{-1}(A)$ is a Nakayama algebra of rank $n + d$ with Kupisch series

$$c(\varepsilon^{-1}(A)) = (\tilde{c}_1, \tilde{c}_2, \cdots, \tilde{c}_n).$$

Note that $\chi(\varepsilon^{-1}(A))$ equals the number of simple components of A.

Recall the following result; see [15,17].

Proposition 5.3. *Let Λ be a cyclic Nakayama algebra and $d \geq 3$. Then Λ is higher Auslander of global dimension d if and only if Λ is isomorphic to*

$$\varepsilon^{-1}(A)$$

where A is a cyclic Nakayama algebra which is higher Auslander of global dimension $d-2$ or isomorphic to

$$\varepsilon^{-1}(A_1 \oplus A_2 \oplus \cdots \oplus A_r)$$

where A_i is a linear Nakayama algebra which is higher Auslander of global dimension $d-2$ for each $i \leq r$.

By Proposition 4.10, we have the following.

Proposition 5.4. *Let Λ be a linear Nakayama algebra and $d \geq 3$. Then Λ is higher Auslander of global dimension d if and only if Λ is isomorphic to*

$$\varepsilon^{-1}(\mathbb{A}_1 \oplus A_1 \oplus \cdots \oplus A_r)$$

where A_i is a linear Nakayama algebra which is higher Auslander of global dimension $d-2$ for each $i \leq r$.

Example 5.5. Let A be a Nakayama algebra with Kupisch series

$$(1, 3, 2, 1, 2, 1).$$

Then $\varepsilon^{-1}(A)$ is higher Auslander of global dimension 3 with Kupisch series

$$(3, 3, 3, 4, 3, 2, 2, 2, 1).$$

Nakayama algebras of global dimension 1 are hereditary, and Nakayama algebras which are higher Auslander of global dimension 2 are the Auslander algebras of Nakayama algebras with radical square zero. For any $d \geq 3$, we can repeatedly apply Proposition 5.4 and Proposition 5.3 to get any connected Nakayama algebra which is higher Auslander of global dimension d.

Example 5.6. Let $n \geq m \geq 0$ and A be a non-simple linear Nakayama algebra. The algebra

$$\Lambda = \varepsilon^{-n}(\mathbb{A}_1^m \oplus A)$$

is a connected Nakayama algebra; it is linear if and only if $n = m$. We have

(1) $\varepsilon^{-m}(\mathbb{A}_1^m \oplus \mathbb{A}_n) \cong N_{mn+n}(n)$ for any $m \geq 1$ and $n \geq 2$;
(2) $\varepsilon^{-m}(\mathbb{A}_1^m \oplus N_3(2)) \cong N_{2m+3}(2)$ for any $m \geq 1$.

Data availability

No data was used for the research described in the article.

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